SANDEEP KUMAR

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PROFESSIONAL SUMMARY

AI/ML Engineer specialising in building and deploying AI-powered applications. MSc Data Science student (graduating Feb 2027) with hands-on experience across the full cycle, from preparing data and training models to launching live web apps. Skilled in Python, PyTorch, Hugging Face, FastAPI, Docker.

EDUCATION

Masters in Data Science
University of Hertfordshire
Hatfield, UK **Sept 2024 - Feb 2027**

Bachelor's in Computer Science
Loyola Academy
Telangana, India **May 2018 - June 2021**

PROJECTS

ScholarAI: document Q&A with page-level citations

Python, FastAPI, FAISS, Hugging Face, Docker | [GitHub] · [Live]

- Built a RAG system from scratch (no LangChain): multi-format ingestion, FAISS retrieval, reranking, and a citation validator that rejects fabricated sources.
- Built an evaluation suite (30 labelled retrieval questions, 26 answer questions): reranking raised MRR from 0.77 to 0.93 on the harder set, and 96% of answers cited the correct page.
- Deployed as a Dockerised FastAPI service with a PDF-preview web UI on Railway, with GitHub Actions CI.

Book Recommender: content-based recommendation engine

Python, scikit-learn, FastAPI, MLflow, Docker | [GitHub] · [Live]

- Built a six-stage TF-IDF and nearest-neighbour pipeline over ~11,000 books, tracked in MLflow and benchmarked against baselines (~5x better than random at genre agreement@5).
- Deployed on Railway with pytest and a CI quality gate that fails builds if model performance drops.

Text Summariser: dialogue summarisation with FLAN-T5

Python, PyTorch, Hugging Face, FastAPI, Docker | [GitHub] · [Live]

- Fine-tuned FLAN-T5 for abstractive dialogue summarisation in a modular training and evaluation pipeline (ROUGE-1 41.2, ROUGE-L 33.5).
- Served via FastAPI in Docker and deployed publicly on Railway.

WORK EXPERIENCE

Data Analyst intern **Dec 2023 – May 2024**
Containe Technologies Ltd.

- Analysed vehicle GPS data (location, speed, idling, movement patterns) in Excel and Power BI to flag unusual activity, including unauthorised trips and suspected device tampering.
- Cleaned and validated tracking datasets and built reports and dashboards for senior analysts, supporting fleet monitoring.

System Associate **July 2022 – Aug 2023**
Infosys Ltd.

- Resolved L1/L2 support tickets and ran routine system checks, backups and maintenance to keep internal business applications and end-user systems running reliably.
- Wrote and executed test cases, fixed bugs and documented processes as part of a software project team, using Java.